PSP-144 · FALSIFIABLE INSTRUMENTATION LAYER · V1.0

AlienEEG-EPF Observation Instrument

The open-source 18/24-channel observation instrument for EPF v3.1 — verified three-chip ADS1299 architecture, EPF-aligned montage, firmware pipeline, and bench-level binding to validation gates EP-1 through EP-6.

Ghost Mesh Research Division

Sovereign Hub · Instrumentation Layer · Reference Build

OCTOBER 2026 · COMPANION TO EPF-3.1 · V1.0 · RESEARCH USE ONLY

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1. Design Charter and Scope

This blueprint specifies the AlienEEG open-source EEG system as the reference observation instrument for the Enhanced Psychics Framework v3.1 (EPF-3.1). It is a complete hardware, firmware, and integration specification: the verified three-chip ADS1299 architecture, the corrected bill of materials, the EPF-aligned 18-channel montage, the on-device processing pipeline, and the binding of every instrument capability to a framework estimator, protocol level, or validation gate. The document absorbs a structured design review in which nine material defects of the original blueprint were identified and corrected, and every corrected value is carried through the rest of the text. Where the original build guide and the framework documentation use different terms for the same object, the framework term wins and the mapping is stated explicitly.

> ***Charter caveat.*** *The instrument is strictly passive: it records electrophysiological signals and injects no current into tissue beyond the microampere lead-off/impedance excitation built into the ADS1299. It makes no medical claims, performs no diagnosis, and is not a treatment device. All framework claims recorded through this instrument remain bound to the EPF-3.1 kill gates EP-1 through EP-6, and the expected-null stance survives unchanged: "psychic" performance, where above-chance transfer is measured, is attributed to subtle-cue inference, statistical anticipation, and inter-person physiological coupling, not to paranormal reception. Research use only.*

18 + 6

populated simultaneous channels, plus 6 hardware spare channels on the daisy chain

1.1 Position in the EPF-3.1 Instrument Hierarchy

EPF-3.1 Chapter 4 restructured acquisition into a two-tier model. The screening tier (S0) is the minimum hardware any operator may use to explore the framework, and the validation tier (V0) is the hardware required before any estimator output may be promoted, registered, or cited. The AlienEEG is designated the V0 reference instrument: its 18 simultaneous channels, auxiliary EMG/EOG inputs, per-channel impedance telemetry, and photonic multi-device synchronization satisfy every V0 entry requirement at component cost near the low end of the tier. The 8-channel OpenBCI path from EPF-3.0 remains a valid S0 screening device, but the v3.1 review made its ceiling explicit: eight channels cannot support a stable inverse estimate or a credible intra- versus inter-network coherence contrast, so screening-tier sessions are structurally incapable of passing gate EP-1 and must be labelled as exploratory in every downstream record.

The designation carries obligations in both directions. The instrument must expose the telemetry the framework's estimators and gates consume, which is why impedance vectors, artifact bitmasks, drop statistics, and pipeline version hashes are first-class fields in the session schema rather than optional logs. Conversely, the framework agrees to retire estimator layers, not the instrument, when a gate fails: a negative EP-3 result retires the dyadic coupling claim, while the recording device remains a competent open-source EEG system with independent scientific value. This separation is deliberate and mirrors the framework's falsifiable posture: the hardware is the witness, never the defendant.

1.2 Scope Boundaries and Non-Goals

Three boundaries keep the project honest. First, there is no stimulation of any kind: transcranial direct-current or alternating-current stimulation, transcranial magnetic stimulation, and any other current- or field-injection modality are excluded from the hardware, the firmware, and the enclosure design, and the expansion port is documented in a way that does not invite them. Second, the device is a research instrument, not a medical device: none of the 24 software functions described in Appendix D may be marketed as diagnostic, and the seizure-detection demonstration is compiled out of framework builds because a seizure alarm is clinical in appearance while remaining unregulated in substance. Third, data sovereignty defaults to the operator: recording, storage, and estimator output remain on the local host unless the operator takes an explicit, granular, revocable sharing action consistent with EPF-3.1 Chapter 9.

Compatibility is claimed against two framework generations. Primary binding is to EPF-3.1 with its six validation gates, its sham-controlled amplification ladder, its imaginary-coherence coupling estimators, and the epf-session/2.0 dashboard schema. Secondary compatibility with EPF-3.0 is retained where operators have not yet upgraded: the instrument streams standard LSL and writes EDF+, so the legacy four-gate program still runs, and Chapter 8 lists the EP-1 through EP-4 bindings that applied under the old gates. In all cases the instrument reports which gate generation a session was executed under, because mixing generations inside one aggregate statistic would quietly corrupt the framework's own accounting.

2. Verified System Architecture and Design-Review Corrections

The verified architecture uses three ADS1299 delta-sigma analog front ends in native SPI daisy-chain mode, providing 24 fully simultaneous channels at 24-bit resolution. Eighteen channels are populated for the EPF-3.1 montage; the remaining six are uncommitted spares available for EOG and EMG references, auxiliary physiological channels, or a future 24-channel upgrade of the same board. Simultaneity is not a convenience feature here: the framework's coupling estimators, phase-locking values, and imaginary coherence all compare phase across channels, so any hardware that time-divides its inputs corrupts the primary measurand before a single line of signal processing runs.

2.1 Signal Chain

Signal flow runs from dry, saline, or sintered Ag/AgCl electrodes through bidirectional ESD protection at the connector, into the three-chip analog front end, then to the microcontroller and the BLE radio, and finally to the host stack that feeds EPF sessions. A single bias drive originates on chip 1 only, and the bias potential also bootstraps the cable shields as an active guard. The chain is battery-powered end to end; the only path to mains potential is the optically isolated charger boundary described in Chapter 8.

Verified signal chain, battery-isolated end to end

```
ALIENEEG-EPF 18-CHANNEL SIGNAL CHAIN (VERIFIED)

  Dry/saline electrodes (18) + A1/A2 mastoid refs
        |
        v
  ESD protection TPD4E1B06 x5 (4-ch each, at connector)
        |
        v
  +-------------------------------------------+
  | Analog front end array (SPI daisy chain) |
  | ADS1299 #1 ch 1-8 [drives BIASOUT] |
  | ADS1299 #2 ch 9-16 |
  | ADS1299 #3 ch 17-24 [2 used, 6 spares] |
  | all 24 channels sampled simultaneously |
  +--------------------+----------------------+
                       | SPI: shared SCLK/DIN/CS,
                       | DOUT(n) -> DAISY_IN(n+1)
                       v
  STM32F411CEU6 Cortex-M4F @100 MHz, FreeRTOS
  + MCP23017 GPIO expander (DRDY/RESET/CS lines)
  DMA SPI reads, IIR notches, artifact flags,
  impedance polling, band-power streaming
                       | UART @ 921600 baud
                       v
  nRF52832 BLE 5.0 module (Raytac MDBT42Q, FCC/CE)
  NUS GATT streaming, 2M PHY, AES-128 CCM
                       |
                       v
  Host: bleak driver -> LSL outlet -> EPF session
  emitter (epf-session/2.0) -> Sovereign Hub bridge
```

$$F_{frame} = 3 \times N_{ch} \text{ bytes per DRDY event } \Rightarrow \text{ 72 bytes at } N_{ch} = 24 \tag{2.1}$$

Equation 2.1 is the payload arithmetic of the daisy chain: each chip contributes three bytes per channel in one shifted transaction, so a full 24-channel frame costs 72 bytes of SPI traffic per data-ready interrupt regardless of how many channels are populated. The BLE packet described in Chapter 5 is a trimmed 60-byte superset of the same frame, adding sequence, status, battery, artifact, and checksum fields. Keeping the two formats related by simple arithmetic makes firmware audits tractable: any discrepancy between captured BLE streams and bench-captured SPI frames is immediately visible as a framing error rather than a silent sample shift.

2.2 Why Three Chips, Not Two Plus a Multiplexer

The original blueprint reached its eighteenth channel by multiplexing an analog input through a TMUX1108 switch. The design review rejected this on four independent grounds, each individually sufficient. First, the switch injects channel-to-channel crosstalk in the -60 to -80 dB range, far below the better than 100 dB channel isolation the ADS1299 provides natively. Second, a multiplexed channel is time-division sampled, so its phase relative to the other channels is an artifact of the switching schedule, which silently invalidates PLV and coherence estimation between that channel and the rest of the montage. Third, electrode switching transients deposit 10 to 100 microvolt settling artifacts into the signal path, an amplitude directly comparable to the cortical signals being measured. Fourth, none of it saved meaningful money: the third ADS1299 adds roughly fifteen dollars to the build and yields six additional simultaneous channels plus a clean expansion story. The corrected design therefore uses three chips unconditionally.

The daisy-chain configuration follows the ADS1299 datasheet exactly. DOUT of chip n connects to DAISY_IN of chip n+1; SCLK, DIN, and CS are shared across all three; and CONFIG1 sets DAISY_EN so the full 72-byte frame shifts out under a single conversion clock. Only chip 1 drives BIASOUT, with the bias amplifiers on chips 2 and 3 disabled through their CONFIG3 registers to prevent multiple low-impedance sources fighting on the bias node. This single-bias discipline matters more than it looks: a divided bias reference would reintroduce exactly the class of common-mode and settling artifacts the three-chip design exists to eliminate, and it would do so selectively, on the channels most likely to be near the disengaged amplifiers.

2.3 Design-Review Corrections Ledger

Before any value in this document was accepted, every numeric claim of the original blueprint was re-derived from component datasheets and vendor pricing. The review found nine material defects; the worst would have produced a non-functional or unsafe build, and several others would have produced quietly wrong expectations about bandwidth, runtime, or cost. Table 1 records each defect and the correction applied. The ledger is retained deliberately: an open-source instrument earns trust by showing its corrections, and the same discipline the framework applies to its own claims applies to the instrument that measures them.

#	Original claim	Defect identified	Correction applied
1	MAX30001 as an 8-channel biopotential AFE option	MAX30001 is a single-channel ECG/BioZ part; no multi-channel EEG capability	Eliminated from consideration; ADS1299 confirmed as the AFE
2	MAX30001 cost efficiency of $1.06 per channel	Based on the false 8-channel premise; true cost is about $4.50 per channel	Comparison removed; ADS1299 at $1.88 per chip-channel is the benchmark
3	Two ADS1299 chips plus TMUX1108 mux for the 18th channel	Mux crosstalk, non-simultaneous sampling, and 10-100 microvolt switching artifacts	Three ADS1299 chips in daisy chain; 24 true simultaneous channels
4	BLE 5.2 on the nRF52832	nRF52832 supports BLE 5.0 only; 5.2 requires nRF52840 or nRF5340	Spec corrected to BLE 5.0; nRF52840 documented as the upgrade path
5	CMRR above 110 dB guaranteed	Datasheet CMRR of 105-115 dB is typical and layout-dependent	Restated as above 100 dB typical, 110-115 dB with optimal layout; layout rules made normative in Chapter 3
6	STM32F411CEU6 priced at $3.80	Current qty-100 pricing is $1.44-$2.20; 48-pin package also short of control lines	Price corrected to about $2.20; MCP23017 GPIO expander added
7	BLE payload of 108 kbps cited with no protocol overhead	ATT/GATT framing and retransmission add 20-30 percent overhead	Bandwidth budgeted at about 156 kbps effective; verified inside 2M PHY capability with margin
8	Battery runtime of 2 hours (original) or 8-10 hours (remix)	System draw near 40 mA at 3.7 V, not 100 mA	Runtime corrected to 20-25 hours on the 1000 mAh pack
9	26.9 channels per $100	Inflated by the MAX30001 and pricing errors above	Recalculated to 19.7 channels per $100 at 18 channels populated

Table 1: Design-review corrections. All corrected values are normative for the rest of this document.

3. Corrected Bill of Materials (EPF-Aligned)

The corrected bill of materials totals approximately $91.30 at quantity-100 pricing, of which the three ADS1299 chips are half. The figure is higher than the original blueprint's estimate because the review replaced optimistic premises with real ones: three analog front ends instead of two, an ESD protection stage the original omitted, a GPIO expander the 48-pin microcontroller genuinely needs, and honest passives and connector budgets. In exchange the build carries 24 hardware channels, a certified BLE module, and a power tree clean enough to hold the CMRR specification in a normal room rather than only on a bench.

$$c_{ch} = BOM / N_{active} \Rightarrow \$5.07 \text{ per channel at 18 channels; } \$3.80 \text{ at 24} \tag{3.1}$$

Equation 3.1 states the cost figure the framework's cost ladder consumes. Against the EPF-3.0 baseline of $15.02 per channel, the corrected design delivers a threefold improvement while roughly doubling channel count, and it holds the V0 tier's intent that validation-grade density be reachable by an individual builder

rather than an institution. Table 2 enumerates the complete build; every line notes the correction or alignment that applies to it, so the table doubles as the procurement checklist.

Function	Part number	Qty	Unit $	Ext $	Notes
Analog front end	ADS1299IPAG (TI)	3	15.00	45.00	Corrected: three chips for 24 true simultaneous channels; no mux
ESD protection	TPD4E1B06DCKR	5	0.40	2.00	Four-channel bidirectional ESD at the electrode connectors
Microcontroller	STM32F411CEU6	1	2.20	2.20	Corrected price; Cortex-M4F, 512 KB flash, 100 MHz
GPIO expander	MCP23017-E/SP	1	0.80	0.80	Added: 16 extra I/O for DRDY, RESET, and CS line management
BLE module	MDBT42Q (nRF52832)	1	4.50	4.50	Corrected: BLE 5.0, 2M PHY, FCC/CE certified module
Charge controller	MCP73831T-2ACI/OT	1	0.50	0.50	USB-C LiPo charging at 500 mA; STAT pin drives indicator LED
USB-C protection	USBLC6-2SC6	1	0.25	0.25	ESD and overvoltage clamp on the charge port
Boost converter	TPS61023DRVT	1	1.20	1.20	3.7 V to 5 V headroom for the analog rail
LDOs (3.3 V)	MIC5219-3.3YM5	2	0.45	0.90	Separate low-noise analog and digital domains
Reference (2.5 V)	REF3325AIDBZR	1	0.60	0.60	Ultra-low-noise reference option for VREFP
Passives	0402 R/C assortment	-	-	3.50	Decoupling, bias networks, input RC filters
Ferrite beads	BLM18PG221SN1D	6	0.10	0.60	Power-domain isolation between analog and digital rails
Electrode connectors	Hirose DF13-20DP-1.25V	2	1.20	2.40	Keyed, locking; two bundles of ten
USB-C connector	TYPE-C-31-M-12	1	0.35	0.35	Charging and firmware offload port
PCB	4-layer ENIG, 80 x 55 mm	1	9.50	9.50	Corrected size for three ADS1299 footprints
Battery	703048 LiPo 1000 mAh	1	3.50	3.50	20-25 h runtime per the corrected draw model
Electrodes	Dry Ag/AgCl button	20	0.50	10.00	18 plus 2 spares; mounted in a silicone cap
Enclosure	3D-printed TPU	1	3.50	3.50	Flexible shell conforming to head curvature

Table 2: Corrected bill of materials at quantity-100 pricing; total approximately $91.30.

3.1 PCB Layout Rules as Part of the Specification

The review's CMRR correction carries a consequence the original blueprint dodged: better than 100 dB common-mode rejection is a property of a layout, not a part number. The following rules are therefore normative, and a board that ignores them is not an AlienEEG regardless of its silkscreen. Analog and digital

grounds join at exactly one star point near the ADS1299 AVSS pins; AVDD and DVDD occupy separate plane regions on layers two and three; a 100 nF decoupling capacitor sits within 0.5 mm of every supply pin on each chip with a 10 microfarad bulk capacitor within 5 mm; and 600-ohm ferrite beads isolate the analog rail from every digital rail that feeds it.

Electrode routing observes equal-length differential pairs with at most 2 mm of mismatch, routed on inner layers between ground planes at a minimum width of 0.2 mm, never beneath digital parts, the switching regulator, or the antenna. A copper guard ring tied to the BIAS-driven shield surrounds the analog inputs, and the TPD4E1B06 protection array sits at the connectors ahead of any trace toward the chips. The SPI bus stays under 50 mm with 22-ohm series dampers at the microcontroller outputs, while the three DRDY lines route directly to interrupt pins without series parts. Finally, the MDBT42Q module sits at the board edge with a 3 mm copper keep-out on all layers beneath its antenna region. These rules, together with the bench gates of Chapter 8, are how the CMRR claim becomes reproducible across independent builders rather than incidental to one lucky prototype.

4. EPF-3.1 Montage Mapping

The 18-channel array covers and extends the EPF-3.0 Chapter 3 montage that EPF-3.1 inherited. The eight core estimator sites remain exactly where the framework placed them, and everything added since exists to make the D3 estimates more trustworthy rather than more numerous. Table 3 assigns every populated channel to a role, and Figure 1 shows the array on the scalp with roles color-coded. The mapping is normative: firmware defaults, impedance gating, and the session schema all assume these assignments, and a build that reassigns channels must say so in its session records or forfeit comparability with the framework's aggregate statistics.

Role group	Sites	Ch #	Function in EPF-3.1
Core D3 estimator sites	Fp1, Fp2, Fz, Cz, C3, C4, Pz, Oz	1-8	Primary estimator anchors: SMR and beta at C3/C4/Cz for the Precision axis; frontal-midline theta at Fz for the Temporal axis; posterior alpha at Pz/Oz for the Boundary axis; frontal sites double as the blink-artifact surveillance set
Estimator-support neighbours	F3, F4, P3, P4	9-12	Neighbour averaging and Laplacian-style spatial filters around the core sites; P3/P4 derivation; improves SMR SNR and stabilizes the posterior coherence geometry
Posterior / temporal auxiliary	O1, O2, T7, T8	13-16	Individualized alpha frequency (IAF) estimation from the occipital grid; auditory and source-monitoring features at T7/T8; additional volume-conduction geometry for the coherence contrast
Linked mastoid reference	A1, A2	-	Linked-mastoid reference (or SRB1 alternative); BIAS return electrode; reference choice recorded per session
Off-montage auxiliaries	EOG-L/R or EMG jaw/neck (bipolar)	17-18	Bipolar EOG (sub-/canthal) or jaw/neck EMG; feeds ICA and ASR clean-up and constitutes the residual-EMG evidence for the gamma gate on the Precision axis

Table 3: Channel assignment for the EPF-3.1 montage.

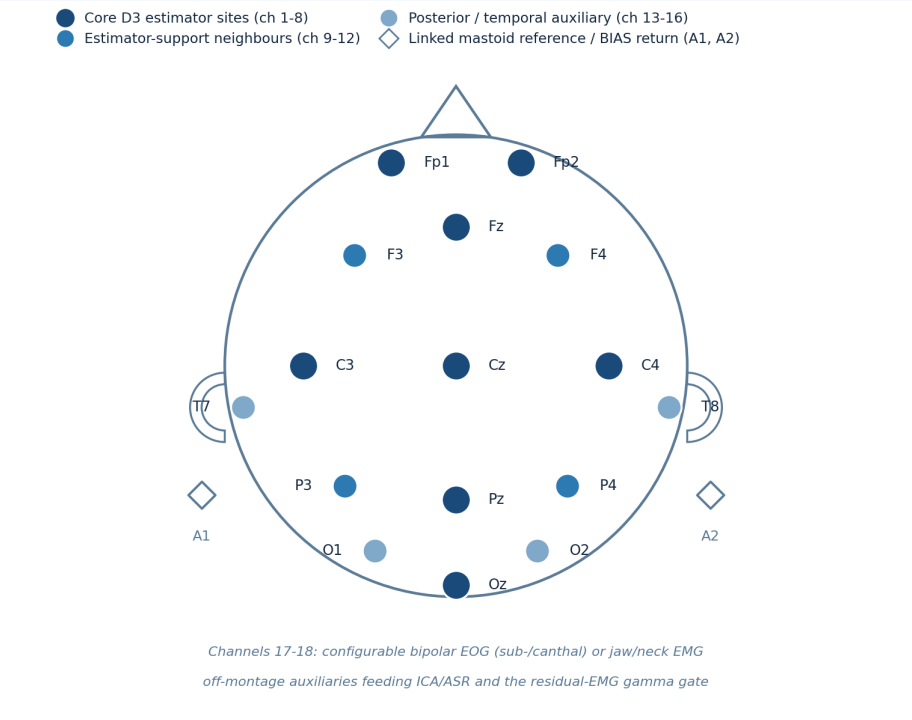


Figure 1: The 18-channel EPF-3.1 montage. Eight core estimator sites (dark) carry the D3 axis anchors; four support neighbours (medium blue) enable spatial filters; four posterior/temporal auxiliaries (light blue) anchor IAF estimation and coherence geometry; diamonds mark the linked mastoid reference and BIAS return; channels 17-18 remain off-montage auxiliaries for ocular and muscle artifacts.

4.1 What the Density Buys, Axis by Axis

For the Precision axis, the review's central finding was that the gamma/alpha ratio is not trustworthy at consumer-grade signal-to-noise because residual jaw and neck EMG survives conventional ICA clean-up. EPF-3.1 responded by re-anchoring the axis on sensorimotor-rhythm and beta features and demoting gamma to exploratory status behind a residual-EMG gate. The AlienEEG supplies exactly what that revision needs: C3, C4, and Cz with F3/F4 and P3/P4 neighbours support variance-ratio spatial filters in the spirit of common spatial patterns, and the bipolar EMG channels provide the quantitative evidence the gate consumes. A gamma-derived estimate is logged only when the auxiliary channels show low muscle residual, and every such log entry carries the EMG statistics that admitted it.

For the Boundary axis, the improvement is geometric rather than cosmetic. EPF-3.1 replaced magnitude-squared coherence with imaginary coherence and the phase-slope index precisely because volume conduction inflates zero-lag synchrony, but the estimator still needs enough posterior channels to distinguish within-network from between-network coupling. Six posterior sites (P3, Pz, P4, O1, Oz, O2) make that contrast meaningful for the first time on a DIY instrument: the intra-parietal-occipital pool can be compared against temporal controls at T7/T8 within a single montage, which is the comparison the B-hat reliability gate (EP-1) actually tests. For the Temporal axis, Fz with F3/F4 neighbours and Fp1/Fp2 surveillance gives frontal-midline theta a defensible spatial definition while keeping the frontopolar channels, which carry the heaviest ocular contamination, out of the theta average unless the EOG regression has passed.

4.2 Reference, Bias, and Electrode Quality Gating

The reference scheme is linked mastoids by default, with the SRB1 register configuration available as an alternative that the session record must declare. BIASOUT drives the common average of the contributing channels and simultaneously bootstraps the cable shields, an arrangement worth 20 to 40 dB of additional common-mode rejection in electromagnetically hostile spaces. Electrode quality is graded by the on-device impedance measurement rather than assumed: below 5 kilo-ohm contact resistance is excellent, 5 to 20 kilo-ohm acceptable, and any core-site reading above 20 kilo-ohm marks the session as gated for estimator promotion. The impedance vector at start, per-block, and end is written into the session record, because a reliable estimate from a deteriorating contact is worse than no estimate at all: it would be indistinguishable from a real effect at exactly the moment the framework can least afford ambiguity.

5. Firmware and On-Device Processing

Firmware is split across the STM32F411, which owns timing-critical acquisition and conditioning, and the nRF52832, which owns transport. The division of labor follows one rule: anything that must not glitch goes on-device, and anything that benefits from auditability goes on the host. The microcontroller runs FreeRTOS with six tasks scheduled around the data-ready interrupt; the radio runs the Nordic SoftDevice with the UART service as its streaming contract. Table 4 defines the task set and Table 5 the conditioning chain, and together they constitute the instrument's real-time specification.

Task	Priority	Period	Function
ADS_SampleTask	Highest	4 ms	Triggered by DRDY interrupt; DMA SPI read of the full 72-byte daisy-chain frame; appends to ring buffer
DSP_FilterTask	High	4 ms	Applies notch and bandpass stages; evaluates amplitude and gradient artifact flags per channel; marks frame status bits
BLE_TxTask	Normal	4 ms	Packs filtered frames into the 60-byte BLE packet; streams over UART at 921600 baud to the nRF52832
LeadOff_Task	Low	100 ms	Polls ADS1299 STATUS lead-off bits; updates per-channel flags and the indicator LED on disconnection
ImpedanceTask	Low	1000 ms	ZAP mode: AC lead-off excitation at 31.25 Hz; measures per-channel impedance; returns to acquisition; logs to the session vector
BatteryTask	Low	10 s	Reads the battery divider; computes state of charge; publishes battery level in the BLE advertisement

Table 4: STM32F411 FreeRTOS task set.

Stage	Type	Parameters	Purpose
1. DC removal	First-order IIR highpass	0.16 Hz	Removes electrode DC offset, which can reach hundreds of millivolts
2. Line notch	Second-order IIR	Q = 35, fc = 50/60 Hz	Power-line interference; 50 Hz variant for European builds
3. Harmonic notch	Second-order IIR	Q = 35, fc = 100/120 Hz	Second harmonic of the line frequency
4. Bandpass	Fourth-order Butterworth IIR	0.5-100 Hz	Physiological band; rejects sub-hertz drift and high-frequency EMI
5. Artifact check	Threshold detector	Flag above 300 microvolts peak-to-peak	Marks blink, jaw, and movement contamination in the status byte
6. Decimation (optional)	2:1 with anti-alias filter	250 to 125 Hz	Halves link load for low-bandwidth feedback sessions

Table 5: On-device conditioning chain (stages apply in order).

5.1 BLE Transport and Packet Format

Each BLE notification carries one sample frame in a fixed 60-byte layout. The design goals are trivial decodability on the host, drop detection without a reliable-link guarantee, and enough status bandwidth that the session record never has to guess about electrode state or battery. The checksum is a plain XOR so a parse failure is attributable in the field with a logic analyzer and nothing more. Frames carry a sequence counter with wraparound, and the host computes drop statistics from gaps, which the session schema records as link quality rather than discarding.

Fixed 60-byte frame; host-side parser is 30 lines of Python

```
BLE packet format (NUS notification, one frame per sample)

  byte 0 frame header 0xA0
  byte 1 sequence number (0-255, wraps; drop detection)
  byte 2 status flags: bits 0-17 lead-off per channel,
              bits 18-20 reserved
  bytes 3-56 18 channels x 3 bytes, 24-bit signed ADC data,
              big-endian two's complement [MSB][MID][LSB]
  byte 57 battery level, percent (0-100)
  byte 58 artifact flags, one bit per channel
  byte 59 checksum: XOR of bytes 0-58

  60 bytes x 250 Hz = 15,000 B/s = 120 kbps raw
```

$$R_{raw} = f_s \times L_{frame} = 250 \times 60\ B = 120\ kbps \tag{5.1}$$

$$R_{eff} = (1 + \eta) \cdot R_{raw},\ \eta = 0.30 \ \Rightarrow\ 156\ kbps\ budgeted \tag{5.2}$$

$$T_{runtime} \approx C_{batt} / I_{sys} = 1000\ mAh / 37\ mA \approx 25\ h\ (specification\ floor\ 20\ h) \tag{5.3}$$

Equations 5.1 through 5.3 are the corrected budgets the review demanded. The raw rate of Equation 5.1 grows by the thirty percent protocol factor of Equation 5.2 to yield the engineering budget actually promised to the radio, and Figure 2a shows that budget against the practical throughputs of the two PHY modes: comfortable margin even on 1M PHY, and a factor of nine on 2M PHY, which absorbs retransmission storms without audible data loss. Equation 5.3 is the runtime model behind the corrected 20 to 25 hour figure; the specification floor is set at 20 hours so a degrading battery announces itself in the telemetry long before a session is threatened. Figure 2b decomposes the 138 milliwatt draw that produces the 37 milliamp estimate.

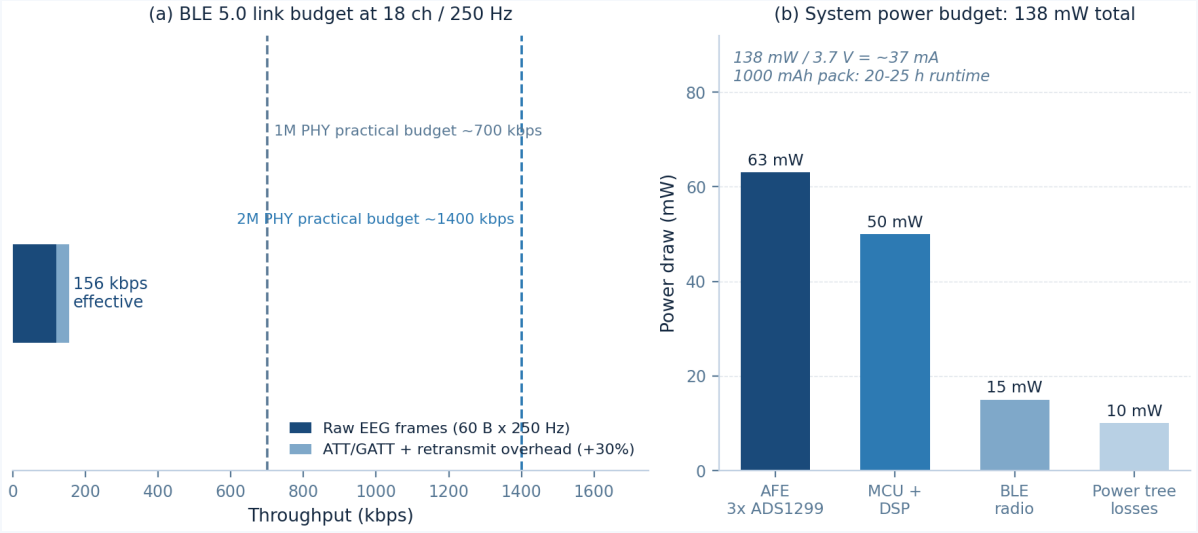


Figure 2: Corrected design budgets. (a) BLE 5.0 link budget: the 156 kbps effective stream (120 kbps raw plus 30 percent protocol overhead) sits far below the practical 1M and 2M PHY budgets. (b) System power budget totaling 138 mW, about 37 mA at 3.7 V, giving 20-25 hours from the 1000 mAh pack.

5.2 Division of Labor: On-Device Versus Host

The device streams band powers on request, but the estimators live on the host by design. On-device work is limited to lossless-critical operations: the conditioning chain of Table 5, impedance measurement, artifact flagging, and the optional low-bandwidth band-power mode for feedback sessions where the radio, not the cortex, is the bottleneck. The host completes ICA and ASR clean-up, imaginary coherence and PLV, MMN extraction, and the regularized inverse that produces the P-hat, B-hat, and T-hat estimates according to EPF-3.1 Equations 5.1 through 5.6, with multitaper spectral option and individualized band edges computed per session rather than fixed.

The split has a governance reason beyond signal quality. The framework now requires that every estimator result be reproducible from a pinned processing environment, so the host pipeline reports a version triple, firmware release, pipeline package version, and container digest, into each session record. Keeping inference on the host keeps that triple meaningful: an on-device estimator would bind scientific provenance to firmware binaries that are harder to audit than a signed Python environment. Band-power streaming for feedback remains the one deliberate exception, and its parameters are declared in the session record whenever the mode is active, including which channels and bands the device computed and with what filters.

Gamma deserves an explicit sentence even though it is off the critical path. The optional band-power stream can carry a gamma band, and nothing prevents a builder from displaying it. What the framework forbids is promotion: any gamma-derived estimate enters the record as exploratory unless the host-side residual-EMG gate, fed by channels 17 and 18, admits it. The instrument's job is to make the evidence for that gate unambiguous, and the firmware therefore never merges the EMG channels into the scalp montage or lets their absence be silently substituted by a default.

6. Integration with EPF v3.1 and the Sovereign Hub

Integration is a contract in three parts: a streaming contract with the local research stack, a schema contract with the framework's session records, and a timing contract with any second headset participating in dyadic protocols. All three are versioned, because the framework's aggregate statistics are only as trustworthy as the weakest producer feeding them. The instrument identifies itself, its firmware, and its configuration in every record it touches, and the hub refuses records that arrive without that identification, which turns accidental misconfiguration into a loud error instead of a silent bias.

6.1 Streaming Contract: LSL Outlets

The host bridge exposes two Lab Streaming Layer outlets. The primary outlet is named AlienEEG, typed EEG, with 18 channels at a nominal 250 Hz in float32, timestamped by LSL with sub-millisecond local jitter. A secondary metadata outlet carries the per-sample status stream, lead-off bits, artifact bitmask, battery, and the periodic impedance vector, so consumers can bind data quality to data without parsing the raw BLE frames. The outlet names and channel labels follow the framework's montage vocabulary exactly, which lets MNE, EEGLAB, BCI2000, OpenViBE, and the OpenBCI GUI consume the stream with no per-lab patching, and lets stimulus presentation software mark events into the same clock for MMN and oddball blocks.

6.2 Schema Contract: epf-session/2.0

Sessions are posted to the local Sovereign Hub bridge at POST /api/v2/sessions using the epf-session/2.0 schema. The instrument contributes the acquisition block shown below: device identity and firmware versions, tier declaration, the montage actually used, the impedance vector at start, per block, and end, link quality from the sequence counters, the auxiliary channel list, and the pipeline version triple. Everything the validation gates will later ask about the recording conditions is captured here, at the time it was true.

epf-session/2.0 acquisition extension for AlienEEG (abridged)

```
POST /api/v2/sessions
{
  "schema": "epf-session/2.0",
  "session_id": "ses_20261019_094231_a41f",
  "acquisition": {
    "tier": "V0",
    "device": {
      "make": "AlienEEG", "rev": "1.0",
      "fw": {"stm32": "1.3.0", "nrf52": "1.3.0"},
      "afe": "3x ADS1299, gain 24", "fs_hz": 250,
      "transport": "ble2m", "phy_budget_kbps": 156
    },
    "montage": ["Fp1","Fp2","Fz","Cz","C3","C4","Pz","Oz",
                "F3","F4","P3","P4","O1","O2","T7","T8",
                "auxEOG_L","auxEOG_R"],
    "reference": "linked_mastoid_A1A2",
    "impedance": {"start": "...", "blocks": ["..."],
                  "end": "...", "core_gate": "pass"},
    "link": {"transport": "ble", "drop_pct": 0.03,
             "jitter_ms": 0.6},
    "aux": ["eog_bipolar", "emg_jaw", "imu_6axis", "ppg_hr"]
  },
  "pipeline": {"fw": "1.3.0", "pkg": "epfpipe 2.4.1",
               "container": "sha256:9f31...c2"},
  "estimates": [ "/* P-hat, B-hat, T-hat with ci90, as v3.1 */" ],
  "safety": {"load_integral": 0.21, "recovery_blocks": 1,
             "flags": ["impedance_warn_P4"]}
}
```

6.3 Timing Contract: Photonic Sync for Dyadic Protocols

Level 4 coupling estimates and gate EP-3 require two headsets to share a clock well enough that phase comparisons mean something. The instrument solves this optically: a phototransistor on each unit watches for a brief light pulse, from an LED wand or a phone screen flash, that both participants can see simultaneously, and each unit stamps the pulse edge against its local clock and broadcasts the timestamp. The host aligns both streams to the shared event. The error budget of Equation 6.1 combines the LED edge uncertainty, below 50 microseconds, with the BLE timestamp quantization of at most one 625-microsecond connection interval, for a combined worst case under one millisecond.

$$\sigma_{sync} \approx \sqrt{(\sigma^2_{LED} + \sigma^2_{ts})} < 1\ ms\ (\sigma_{LED} < 50\ \mu s;\ \sigma_{ts} \leq 625\ \mu s) \tag{6.1}$$

One millisecond is deliberate: imaginary coherence at the individual alpha frequency tolerates timing error an order of magnitude smaller than the period of the band being compared, so sub-millisecond sync protects the estimator rather than merely decorating it. Where a lab needs tighter margins, the documented upgrade path swaps the radio module for an nRF52840, whose 802.15.4 radio supports OpenThread time synchronization at roughly 100-microsecond effective error. The protocol logs the achieved sync error of every dyadic session into the record, because a coupling lift measured on a sloppy clock is exactly the kind of

artifact EP-3 exists to catch.

6.4 Dashboard Hooks and Demand Control

The hub binds the live P-hat, B-hat, and T-hat estimates to the PSP-144 axis monitors, and Crucible progress tracking consumes the ladder telemetry of Chapter 7 directly. Two governance hooks from EPF-3.1 Chapter 10 are wired at the instrument level rather than bolted on later. Demand control reads the session caps, daily and weekly, that the framework imposes, and the device telemetry feeds the load integral so caps are computed from exposure rather than trust. And the research dashboard is structurally separated from any public facing claim surface: the instrument publishes estimates to the local hub only, and nothing in its firmware can post to external services, which enforces the framework's position that a dashboard is a mirror for the operator, not a megaphone.

7. Amplification Ladder and Protocol Support

The five-level amplification ladder belongs to EPF-3.1 Chapter 6, and its semantics, including the rule that a yoked sham arm accompanies every training level from the first cohort, are owned there. The instrument's job is to supply each level's metrics with known provenance and to make the sham condition structurally possible. The BLE protocol therefore carries a sham-mode flag: when it is set, the feedback stream delivered to the operator is decoupled from the operator's own EEG and driven by a recorded replay or a contingency-stripped signal matched in bandwidth and amplitude. The session record always shows which mode ran, so a sham-arm session can never masquerade as a contingent one in any aggregate.

Level	Objective (owned by EPF-3.1 Ch. 6)	Instrument-supplied metrics	Sham provision
L1	Baselines, IAF estimation, MMN and oddball blocks	Eyes-open and eyes-closed baselines; occipital IAF with credible range; MMN at Fz with T7/T8 references; LSL stimulus markers time-locked to samples	Not applicable; L1 is measurement only
L2	SMR up-training at Cz with theta monitoring	On-device SMR (12-15 Hz) and theta band powers streamed at 2 Hz; host-side SMR z-scored to the operator's own L1 baseline; CUSUM stop telemetry	Yoked sham mandatory: replay of a matched prior session drives the same feedback display
L3	Alpha-theta crossover with IAF-centred bands	Alpha (IAF-4 to IAF+2 Hz) and theta (4-8 Hz) band powers; crossover event flags; dissociation telemetry feeding the recovery-block rule	Yoked sham from the first cohort; CADSS checkpoint prompts logged as session markers
L4	Dual-headset alpha-band coupling	Imaginary coherence as primary with PLV logged alongside, both at the dyad's mean IAF; photonic sync error budget recorded per session	Coupling compared against shuffled-pair and positive-control subtle-cue conditions; both run on identical instrument configuration
L5	Sealed blind trials of the Channel/Remote-View protocol	Raw 60-byte frames written to sealed storage; cryptographic target and transcript hashes chained into the session record; N = 200 registration export	Protocol-owned blinding; instrument guarantees the log is append-only and hash-chained

Table 6: Per-level metric supply and sham provision across the amplification ladder.

Safety telemetry rides the same paths as the science. The load integral defined by EPF-3.1 Equation 6.1 is computed on the host from session duration, artifact fraction, boundary-exposure flags at level three, and recovery blocks, and the device contributes the raw ingredients rather than a pre-baked score, keeping the safety arithmetic auditable in the same place as the estimators. Automatic stop rules are wired to instrument signals: an estimator that diverges from its own uncertainty bound, an impedance failure on a core site, or an artifact storm that survives clean-up all trigger a controlled end of session with the reason recorded. The intent is that an operator following the dashboard alone still ends a session the framework would have ended, which is what a safety interlock means in a system whose users are also its investigators.

8. Validation, Safety and Falsifiability Binding

An instrument that feeds a falsifiable framework must itself be held to bench-level falsification: numeric pass criteria, measured the same way by every builder, before any human wears it. Table 7 is that gate. All nine tests are executed on every assembled unit, in order, and a unit that fails any line is quarantined until it passes; the biological checks that follow assume the bench gates. The criteria reproduce the corrected performance table of Appendix A, so a unit passing Table 7 has demonstrated, on itself, the specifications the rest of this document claims.

Bench test	Method	Pass criteria
Input noise floor	Short all inputs (INxP to INxN); record 30 s at 250 Hz, gain 24; compute RMS per channel	Below 1.0 microvolt RMS per channel
Common-mode rejection	Inject 60 Hz, 1 Vpp on both inputs via a balanced divider; compute 20 log(Vcm/Vout)	Above 100 dB (110-115 dB expected with optimal layout)
Channel crosstalk	Drive 50 Hz, 100 microvolts pp on channel 1; measure all other channels	Below -80 dB isolation between channels
SPI integrity	Oscilloscope on SCLK, MOSI, MISO at 4 and 20 MHz	Clean edges, rise time below 5 ns, no ringing above 10 percent
BLE range and drops	Pair with host; walk to 10 m with obstacles	Stable connection; packet loss below 0.1 percent
Battery runtime	Full charge; 18 channels at 250 Hz with BLE active; run to cutoff	Above 18 hours to cutoff voltage
Lead-off detection	Disconnect one electrode during recording	Correct channel status bit asserted within 1 s
Reference voltage	Measure VREFP on each ADS1299	4.490-4.510 V (internal reference) or 2.495-2.505 V (REF3325)
System current	DC current meter on the battery output	Below 50 mA total at 3.7 V

Table 7: Bench validation gates; every assembled unit must pass all nine before human use.

Biological validation then confirms the instrument on a person rather than a simulator. Posterior alpha is verified at O1/O2 by the eyes-closed versus eyes-open contrast appearing within five seconds; motor rhythm is verified at C3/C4 by event-related desynchronization during imagined hand movement; the blink test confirms that deliberate eyeblinks deposit their characteristic 100 to 300 microvolt signature in the frontal channels while sparing posterior channels; and a line-noise FFT confirms the notch chain reduces mains interference below one microvolt RMS. A unit passing the bench table but failing a biological check has a configuration or montage fault, not a silicon fault, and the failure is recorded against the affected channels in the unit's validation notebook.

8.1 Binding Instrument Capabilities to EP-1 through EP-6

The framework's six validation gates ask different questions, and each one presupposes an instrument capability that can be pointed to in this document. Table 8 makes the presupposition explicit in both directions: which gate draws on which instrument property, and what happens when the gate fails. The consequence column implements the charter's separation of witness and defendant: gate failure retires the corresponding claim layer, never the instrument, and never the underlying axis algebra that merely consumes its outputs.

Gate	Question asked	Instrument capability relied upon	On gate failure
EP-1	Are estimator features reliable enough to carry weights?	V0 density (Table 3), impedance telemetry and gating, per-session IAF; screened S0 data excluded from promotion	Estimator demoted to exploratory; instrument continues as recorder
EP-2	Does neurofeedback beat a sham-controlled arm?	SMR band streaming at 2 Hz, yoked-sham mode flag, CUSUM stop telemetry	Ladder level retired; sham arm published alongside
EP-3	Does dyadic coupling exceed shuffled and positive controls?	Photonic sync with Equation 6.1 budget logged; ImCoh primary, PLV logged; identical config across conditions	Coupling claim retired; sync artifacts ruled out first
EP-4	Does the trial layer beat chance under full blinding?	Sealed append-only raw logging; target and transcript hashes chained into the record	Trial layer retired per the expected-null clause
EP-5	Do results replicate outside the originating lab?	Version triple (firmware, pipeline, container) in every record; public BOM and register map; BIDS export	Non-replicated estimators lose operative status framework-wide
EP-6	Can isolation and scoring be defeated by a red team?	Environment and aux sensor logging; scoring inputs recorded raw; isolation conditions quantified per session	Defeated channels published; protocol revised before retest

Table 8: Binding of EPF-3.1 validation gates to instrument capabilities and failure consequences.

8.2 Safety Envelope

The safety envelope matches EPF-3.1 Chapter 9 point for point. Power is battery-isolated from mains: the only galvanic path to the grid is the charge port, and the firmware refuses to stream while USB power is present, making charge-and-record simultaneously impossible rather than merely discouraged. Electrodes are passive by construction; the only current the device can source into tissue is the microampere AC lead-off excitation used for impedance measurement, which is part of the ADS1299's designed operating envelope.

Session-length caps and the recovery-block rule surface as telemetry the operator cannot silently dismiss, and adverse-event flags are first-class fields in the session schema so the incident registry of the framework's governance chapter is fed by the instrument rather than by memory.

Two explicit non-claims close the chapter. The device performs no clinical monitoring of any kind, and the seizure-detection function demonstrated in the upstream project is compiled out of framework builds because a feature that looks medical must either be held to medical-device discipline or not ship. And nothing in this document asserts that the instrument detects, amplifies, or validates any paranormal ability; it is a competent, honest, open EEG recorder whose scientific job, in this framework, is to make the null hypothesis work for a living.

9. Open-Source Deliverables and Reproducibility

Every artifact needed to build, verify, and operate the instrument is released. The repository layout below is the release contract: a tag that lacks any element of it does not count as a version. Hardware design files and this blueprint are published under CC BY-SA 4.0; firmware repositories carry their own license files derived from the upstream SDK example terms. Reproducibility discipline mirrors the framework's containerization rule: releases are signed, the firmware version triple is written into every session record, and the bench validation notebooks of Chapter 8 run from the same tag they validate.

Release contract: one tag, everything above, signed

```
alieneeg/
  hardware/
    kicad/ schematic + 4-layer PCB (star ground,
                     guard ring, driven shield, antenna keep-out)
    BOM.csv Table 2 as machine-readable manifest
    assembly.md Appendix C sequence with per-phase gates
  firmware/
    stm32/ FreeRTOS tasks (Table 4), ADS1299 driver,
                     CMSIS-DSP chain (Table 5), DFU bridge
    nrf52/ nRF5 SDK 17 + SoftDevice S132, NUS GATT,
                     AES-128 CCM, DFU bootloader
  host/
    bleak_driver/ BLE transport, 60-byte frame parser
    lsl_bridge/ EEG + status outlets (Ch. 6.1)
    session_emitter/ epf-session/2.0 POST /api/v2/sessions
    mne_import/ MNE-Python reader + BIDS-EEG exporter
  validation/
    bench/ Table 7 notebooks, one per gate
    biological/ alpha, mu, blink, line-noise checks
    ep_gates/ EP-1..EP-6 validation notebooks
  docs/
    register_map.md Appendix B as living document
    safety.md Chapter 8.2 envelope + incident registry
```

This layout is what makes gate EP-5 reachable at all. An independent laboratory that clones a tag receives the same PCB files, the same firmware, the same bench gates, and the same importers as the originating build, which reduces replication friction to the honest minimum: the cost of building a unit and running a cohort, with no hidden configuration debt. The MNE and BIDS exporters exist so that replication data can flow into public archives under the consent rules of EPF-3.1 Chapter 9, granular, revocable, and anonymized at the point of export rather than by later promise. An instrument that cannot be independently rebuilt cannot participate in a replication gate, and this repository layout is the instrument's side of that bargain.

Appendix A: Corrected Performance Specifications

Table 9 consolidates the review's numeric corrections into one reference, comparing the EPF-3.0 baseline instrument, the original blueprint claims, and the corrected values this document carries. The corrected column is the specification of record; where a corrected value depends on build discipline, the dependency is stated rather than hidden, because a spec that fails silently in the field is worse than an honest range.

Parameter	EPF-3.0 baseline	Original blueprint	Corrected value
Channels	8	18 (16 + mux)	18 populated of 24 simultaneous
Input noise	0.948 microvolt RMS claimed	Below 1.0 microvolt pp	~0.25 microvolt RMS at gain 24, 250 SPS
Resolution	16-bit	24-bit	24-bit (confirmed)
CMRR	96 dB claimed	110 dB guaranteed	Above 100 dB typical; 110-115 dB with optimal layout
Sampling	250 Hz	500 Hz optional	250 Hz default; 500 Hz tested; 16 kSPS maximum
Bandwidth	0.5-45 Hz	0-100 Hz	0.5-100 Hz with 0.16 Hz DC highpass
Wireless	BLE 4.2	BLE 5.2 claimed	BLE 5.0 (nRF52832); 5.2 via nRF52840 upgrade
Battery life	2 hours	8-10 hours	20-25 hours (draw model ~37 mA)
BOM cost	$60.10	$66.90	$91.30 (three-chip corrected)
Cost per channel	$15.02	$3.72	$5.07 at 18 ch; $3.80 at 24
MCU	STM32F103, 64 KB	STM32F411	STM32F411, 512 KB, plus MCP23017 expander
Link budget	not stated	108 kbps raw only	156 kbps effective incl. 30% overhead
Runtime at spec	not stated	not stated	18 h floor to cutoff; 20-25 h typical

Table 9: Performance specification lineage; the corrected column is authoritative.

Appendix B: ADS1299 Register Quick Reference

Table 10 lists the register configuration the reference firmware applies to each ADS1299. Values are for chip 1 unless noted; chips 2 and 3 differ only in CONFIG3, where the bias amplifier is disabled so that a single bias drive exists on the chain. The register map is also published as a living document alongside the driver source, and any deviation in a builder's firmware must be declared in the session record for the reasons provenance matters.

Register	Addr	Key bits	AlienEEG setting
ID	0x00	REV_ID[2:0]	Read-only; returns 0x3E for ADS1299
CONFIG1	0x01	DAISY_EN, CLK_EN, DR[2:0]	DAISY_EN = 1 (daisy chain); DR = 110 (250 SPS)
CONFIG2	0x02	INT_CAL, CAL_AMP, CAL_FREQ	0x00 normal operation; 0x50 for test-signal calibration
CONFIG3	0x03	PD_REFBUF, BIAS_EN, BIASREF_INT	0xEC on chip 1: internal reference buffer plus bias drive; bias disabled on chips 2-3
LOFF	0x04	COMP_TH, ILEAD_OFF, FLEAD_OFF	AC lead-off excitation at 31.25 Hz for ZAP impedance mode
CH1SET-CH8SET	0x05-0x0C	PD, GAIN[2:0], MUX[2:0]	0x60: gain 24, normal input, channel powered
BIAS_SENSP/N	0x0D/0x0E	BIASP/N[7:0]	All 8 channels per chip contribute to the bias average
LOFF_SENSP/N	0x0F/0x10	LOFFP/N[7:0]	All channels enabled for lead-off detection
GPIO	0x14	GPIOC, GPIOD	Inputs by default; status signaling
MISC1	0x15	SRB1	SRB1 = 1 optional: SRB1 pin as reference for all channels

Table 10: Register quick reference for the three-chip chain.

Appendix C: Assembly, Bring-Up and Biological Validation

Assembly proceeds in five phases, and the gate column of Table 11 is the whole point: no phase begins until the previous phase's gate passes, because debugging a three-chip analog chain with a suspect power tree wastes boards and teaches the wrong lessons. The ADS1299 packages deserve special respect; their 0.5 mm pitch rewards a paste stencil and punishes impatience, and a single solder bridge on a supply pin can masquerade as a dozen different faults. Builders without reflow experience should budget a sacrificial

practice board and treat the microscope inspection of step 23 in the full build guide as the gate it is.

Phase	Scope	Gate to pass before next phase
1. Power	Charge controller, boost, both LDOs, reference, decoupling, ferrites	USB-C input 4.9-5.1 V; DVDD and AVDD 3.28-3.32 V; reference 2.495-2.505 V; no load current anomaly
2. Control	STM32F411, MCP23017, bypass caps, SWD header	LED-blink firmware runs; MCP23017 acknowledges at I2C address 0x20
3. Analog front end	Three ADS1299 chips by reflow; inspection under magnification	RREG of ID register returns 0x3E on all three chips at SPI 1 MHz debug clock; continuity on supplies and inputs
4. Radio and connectors	MDBT42Q, electrode and USB-C connectors, battery JST	nRF52832 firmware flashed over SWD; VDDMAIN 3.3 V; NUS advertisement visible to nRF Connect
5. Integration	Electrode cables with BIAS-driven shields, enclosure, battery	Full Table 7 bench sequence passes; charge indicator functional; no component contact with enclosure

Table 11: Five-phase assembly sequence with per-phase gates.

Biological validation completes bring-up on a willing colleague rather than on hardware. The four checks of Chapter 8, posterior alpha modulation, motor-rhythm desynchronization, blink containment, and line-noise rejection, are run in one sitting and recorded in the unit's validation notebook with the affected channels noted. A unit that passes Table 7 and all four biological checks is enrolled in the framework as a V0-tier instrument, its tag recorded in future session metadata; a unit that fails a biological check remains a competent general-purpose recorder but is not eligible to produce promotion-eligible estimates until the fault is found.

Appendix D: 24-Function Catalogue and EPF-3.1 Dispositions

The upstream project advertises twenty-four software functions. A framework that audits its estimators has no business importing a feature list unexamined, so Table 12 disposes each function against EPF-3.1: operative means it serves a ladder level, gate, or governance rule as built; exploratory means it is supported but its outputs never reach promotion paths; out of scope and excluded mean the framework either has no use for it or prohibits it in builds. The dispositions are normative for framework builds and may be revisited only through the same revision process as any other claim.

#	Function	Category	EPF-3.1 disposition
1	Adaptive common spatial patterns (on-device)	Signal processing	Operative: spatial filters for SMR features; pre-trained W loaded from the operator's own L1 data
2	Welch band-power streaming	Signal processing	Operative: low-bandwidth feedback path for L2/L3; parameters declared in session record when active

#	Function	Category	EPF-3.1 disposition
3	Online artifact subspace reconstruction (ASR)	Signal processing	Operative: mandatory pre-ICA clean-up per EPF-3.1 Ch. 5; k = 5 default logged
4	FastICA / infomax (host-side)	Signal processing	Operative: ocular and cardiac removal using aux channels 17-18
5	Phase-amplitude coupling (modulation index)	Signal processing	Exploratory: CFC formalization with surrogate controls; never promotion-eligible
6	Inter-channel coherence / synchrony index	Signal processing	Operative: B-hat network contrast; imaginary coherence computed host-side as primary
7	ZAP per-channel impedance mapping	Hardware	Operative: session gating and the impedance vector in the record
8	Active electrode driven shield	Hardware	Operative: 20-40 dB common-mode improvement; part of layout spec
9	6-DOF IMU motion compensation	Hardware	Partial: recommended; regression residuals must not leak into estimator inputs (documented constraint)
10	PPG heart-rate sync channel	Hardware	Operative: cardiac artifact removal; HRV as workload covariate
11	BLE DFU firmware update	Hardware	Operative: signed releases; version pinning depends on it
12	GSR auxiliary channel	Hardware	Exploratory: arousal covariate; consumes a spare channel; declared in record
13	SSVEP frequency-tagging decoder	BCI application	Out of ladder scope; retained as a stim-clock validation aid
14	P300 speller pipeline	BCI application	Out of ladder scope; not used by framework protocols
15	Neurofeedback gamification app	BCI application	Operative for L2/L3 feedback UI; sham-mode flag mandatory in framework builds
16	Automatic sleep staging	BCI application	Exploratory: outside framework scope; standalone use only
17	Native LSL integration	Infrastructure	Operative: primary integration path (Ch. 6.1)
18	Encrypted BLE (AES-128 CCM, LE Secure Connections)	Infrastructure	Operative: required for any shared or archived session data
19	EDF+ recording with annotations	Infrastructure	Operative: archival format; BIDS conversion provided in host tools
20	On-device seizure detection	BCI application	Excluded: not a medical device; compiled out of framework builds
21	MQTT cloud telemetry bridge	Infrastructure	Partial: allowed only after anonymization and explicit consent; differential privacy deferred to backlog
22	Sinter-bonded dry electrode array with grading	Hardware	Partial: A/B/C impedance grading aligns with tiering; electrode validation pending field data
23	Modular 32-channel expansion port	Hardware	Deferred: activation trigger is the funded 32-channel validation cohort of the EPF-3.1 backlog
24	Photonic clock synchronization	Hardware	Operative: EP-3 critical path; sync error logged per dyadic session

Table 12: All 24 upstream functions with normative framework dispositions.

D.1 Recommended Open-Source References

- OpenBCI Cyton schematics (github.com/OpenBCI/Docs): reference ADS1299 layout from a shipping open-hardware device.
- ADS1299EEGFE-PDK evaluation package (ti.com): TI's official front-end reference design for power and input networks.
- BEATS 32-channel EEG (arxiv.org/abs/2203.02102): daisy-chain configuration precedent with STM32 firmware.
- MNE-Python (mne.tools): ICA, spectral, connectivity, and BIDS export stack used by the host pipeline.
- BCI2000 (bci2000.org) and OpenViBE: real-time frameworks consuming the LSL outlet for paradigms beyond the ladder.
- pyEDFlib (pypi.org/project/pyEDFlib): EDF/EDF+ archival writer used by the session recorder.
- CMSIS-DSP (github.com/ARM-software/CMSIS-DSP): ARM-optimized filter primitives behind the Table 5 chain.
- nRF5 SDK 17.1.0 (infocenter.nordicsemi.com): SoftDevice S132, NUS service, DFU bootloader, and BLE security.